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(54) **SOUND-GENERATING TIMEPIECE**

(71) Applicant: **PATEK PHILIPPE SA GENÈVE**, Geneva (CH)

(72) Inventors: **Quentin Benoist**, Grand-Lancy (CH); **Sylvain Geiser**, St-Cergue (CH); **Eric Le Gall**, Nyon (CH); **Alessandro Vandini**, Tannay (CH)

(73) Assignee: **PATEK PHILIPPE SA GENÈVE**, Geneva (CH)

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G04B 21/08 (2006.01)
(Continued)

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(58) **Field of Classification Search**
None
See application file for complete search history.

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Primary Examiner — Renee S Luebke
Assistant Examiner — Matthew Daniel Hwang
(74) *Attorney, Agent, or Firm* — NIXON & VANDERHYE

(57) **ABSTRACT**

Disclosed is a mechanical wristwatch including a case having a middle, a back cover and a bezel, a vibration-generating device for generating vibrations which are intended to produce at least one sound, which is accommodated in the case, a vibration-amplifying for amplifying the vibrations generated by the vibration-generating device, and a membrane arranged to receive the amplified vibrations and generate a sound from the inside to the outside of the wristwatch. The vibration-generating device is a rotary element having a reading surface including at least one groove arranged to generate sounds. The wristwatch also includes a unit for rotationally driving the rotary element, an actuator for such unit, and a transmission member arranged
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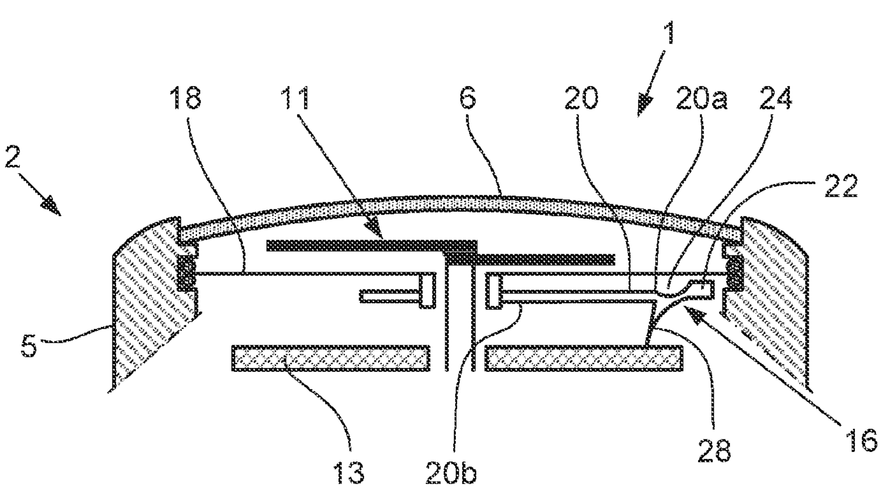

to cooperate with the rotary element to transmit, to the vibration-amplifying device, the vibrations generated by the groove during rotational movement of the rotary element.

24 Claims, 6 Drawing Sheets

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G04B 23/02	(2006.01)
G04B 37/00	(2006.01)

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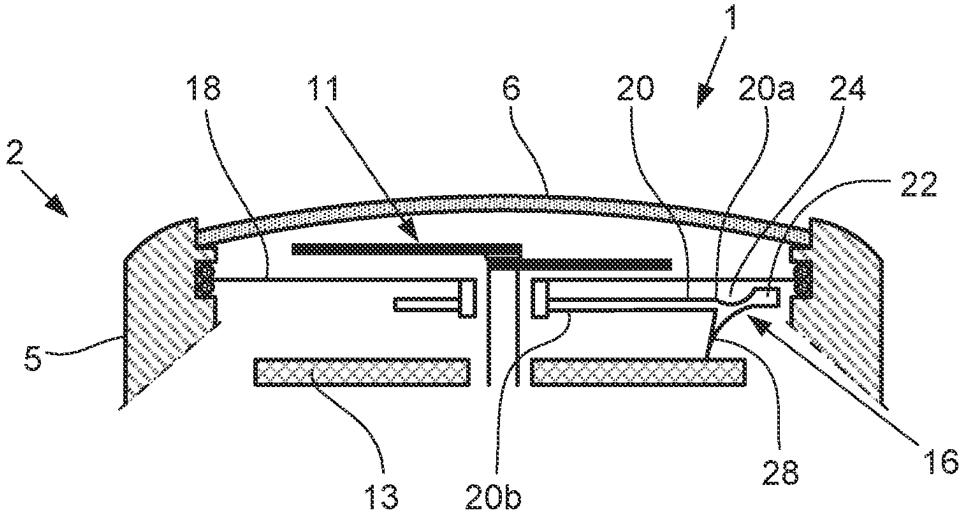

Fig. 1

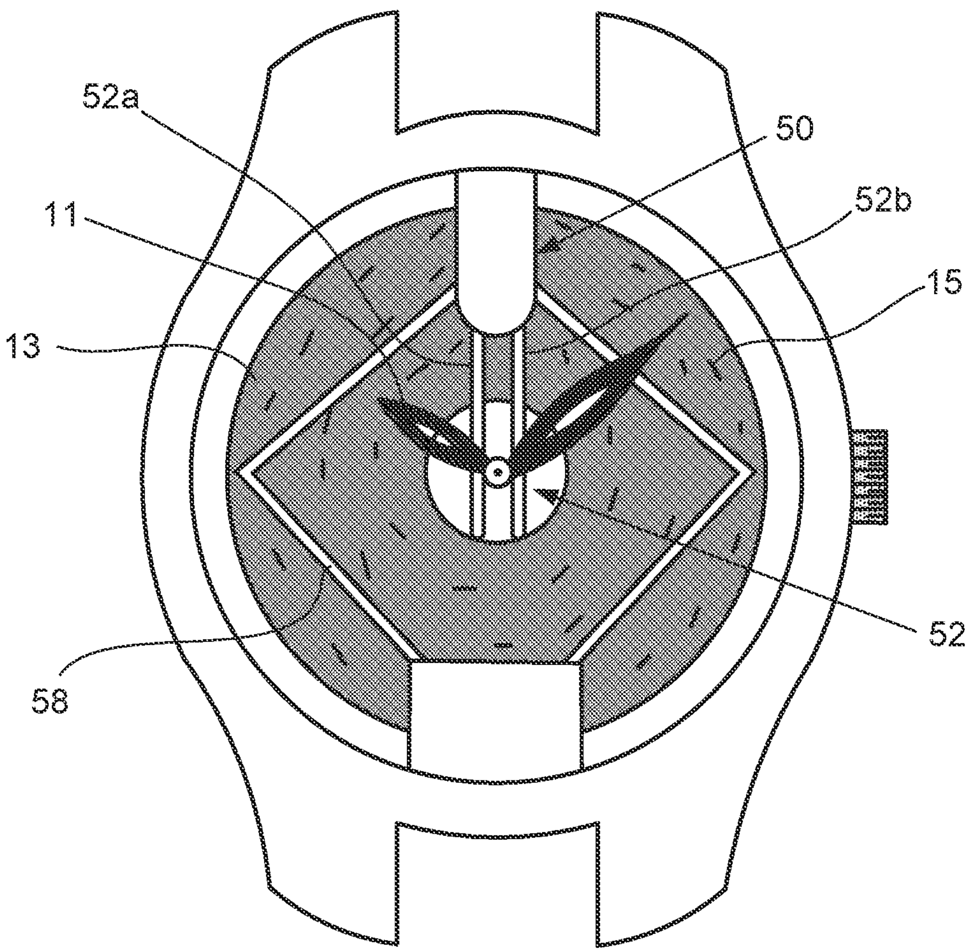

Fig. 2

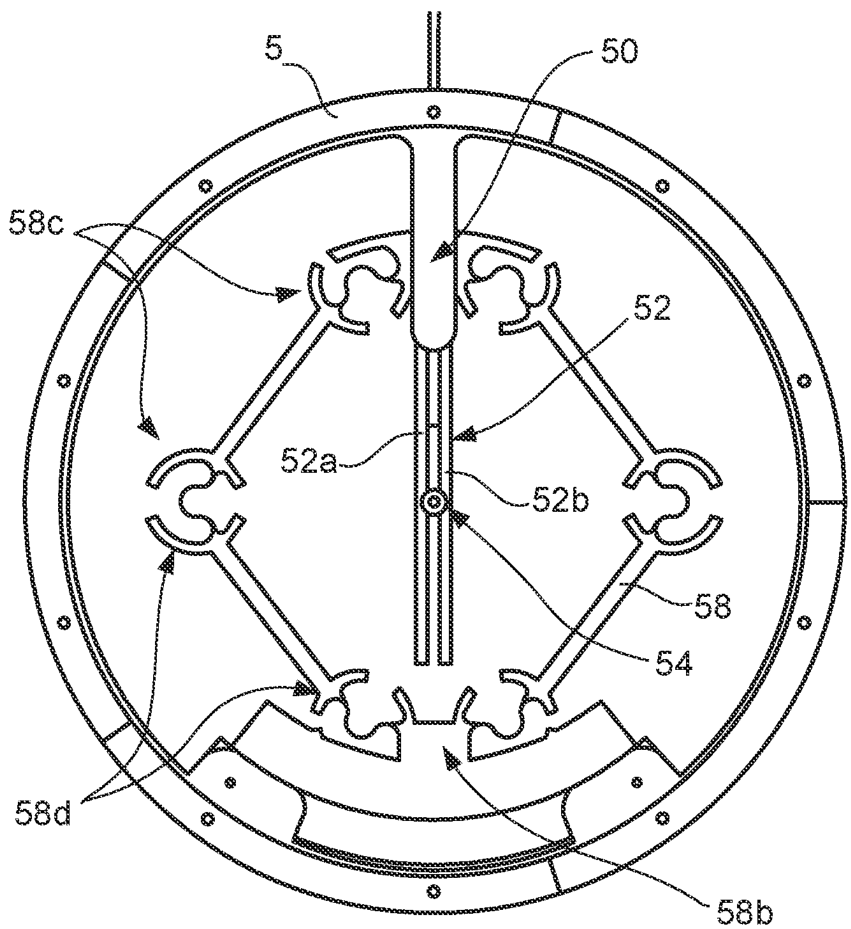

Fig. 3

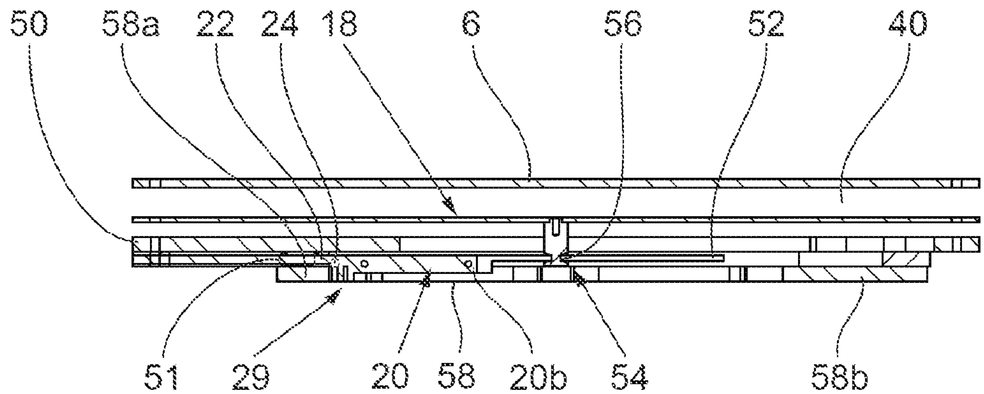

Fig. 4

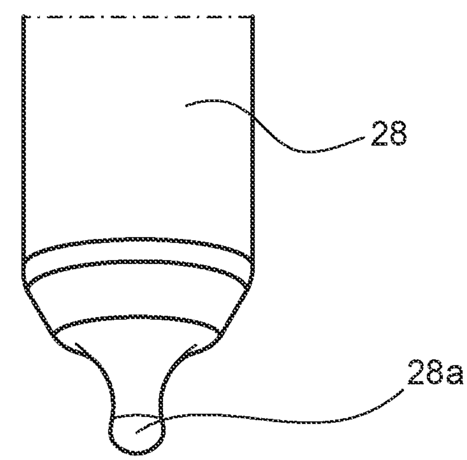

Fig. 5

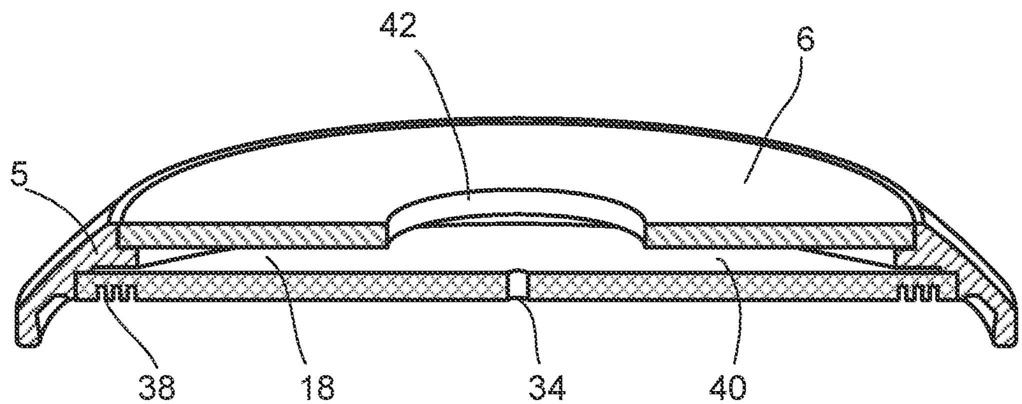

Fig. 6

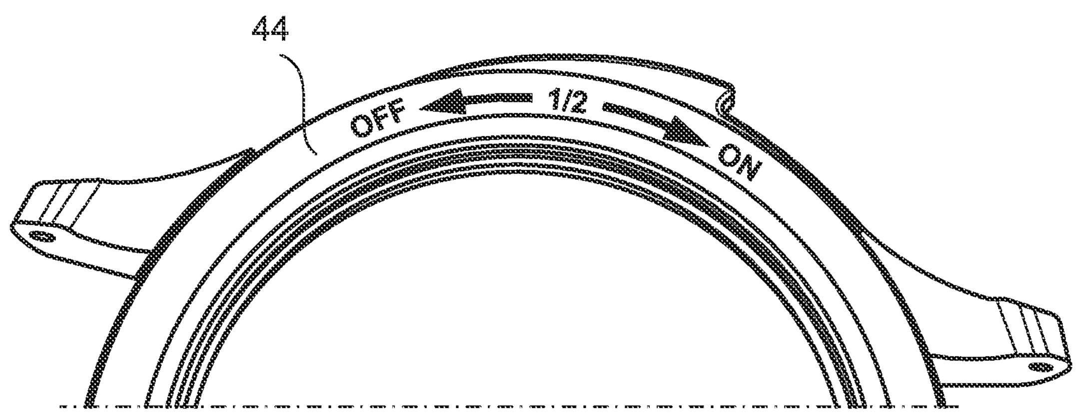

Fig. 7

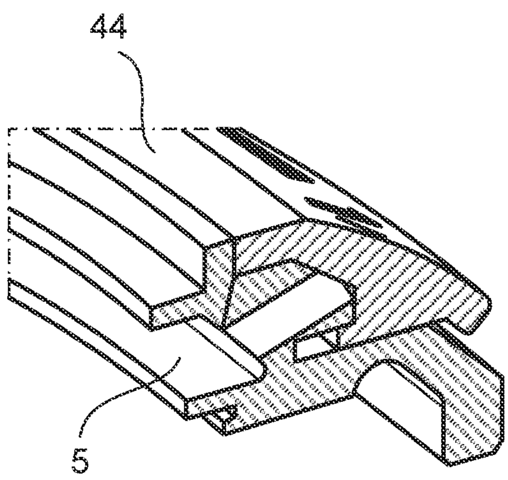

Fig. 8

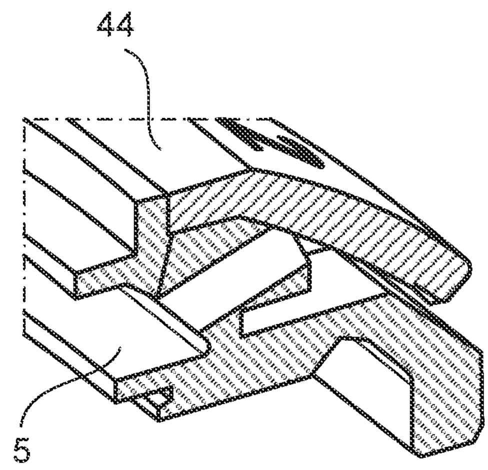

Fig. 9

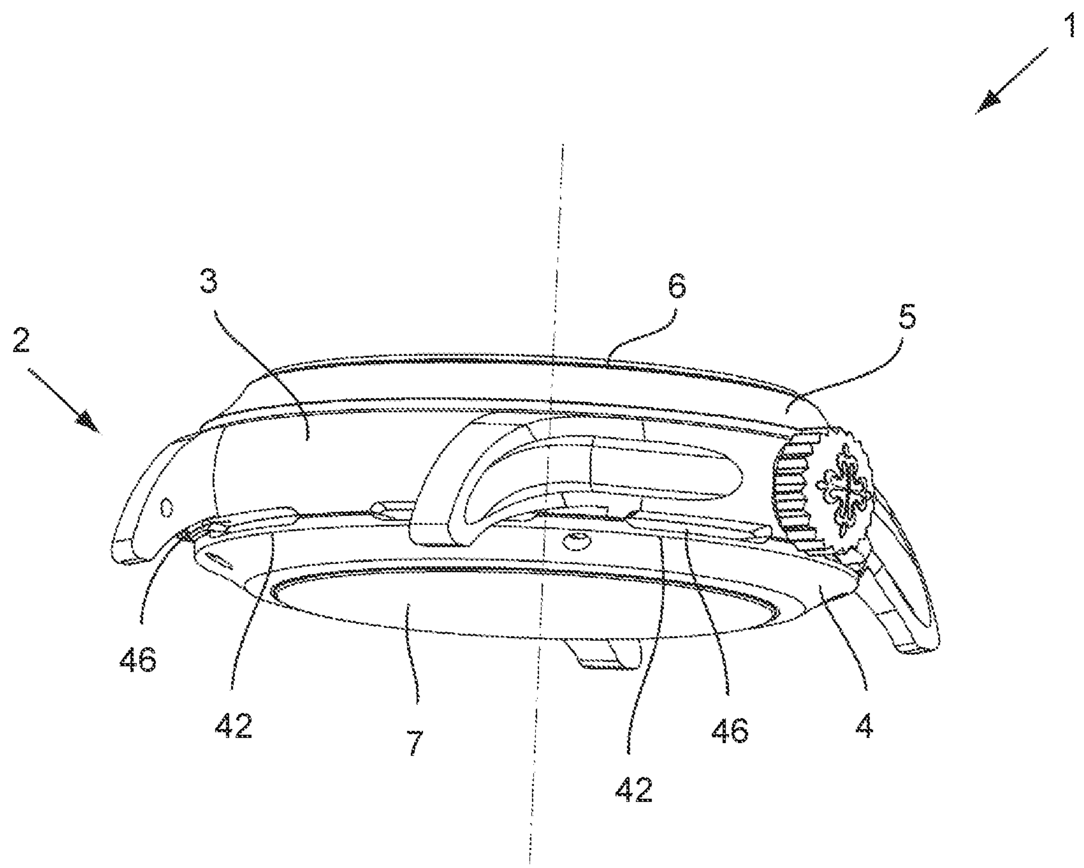

Fig. 10

SOUND-GENERATING TIMEPIECE

This application is the U.S. national phase of International Application No. PCT/EP2020/079955 filed Oct. 23, 2020 which designated the U.S. and claims priority to EP 19205441.9 filed Oct. 25, 2019, the entire contents of each of which are hereby incorporated by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention relates to a mechanical timepiece, and more specifically a mechanical wristwatch, comprising a case having a middle, a back cover and a bezel, a device for generating vibrations which are intended to produce at least one sound, which is accommodated in said case, a device for amplifying the vibrations generated by said vibration-generating device, and a membrane arranged to receive said amplified vibrations and generate a sound from the inside to the outside of the timepiece, and more specifically of the wristwatch.

In the present description, the expression "mechanical timepiece or wristwatch" means that the timepiece or wristwatch comprises only mechanical elements for performing its different functions, and more particularly that the sound and the amplification thereof are produced solely using mechanical elements.

Description of the Related Art

Such a timepiece is for example a wristwatch with a striking-mechanism (such as a minute-repeater, grande sonnerie, petite sonnerie, alarm) for which the vibration-generating device is a gong arranged to be struck by a hammer when the striking-mechanism is engaged to generate vibrations in order produce a sound. A watch with a striking-mechanism generally comprises two gongs, one to produce a deep sound and the other to produce a high-pitched sound. Even with modifications made to the two gongs and to the alternation with which they strike the hours differently from the quarter-hours or minutes, the watch with a striking-mechanism is limited to only two different sounds corresponding to each of the two gongs.

Furthermore, a pocket watch is known allowing the time to be spoken, as described in patent CH 4918. The pocket watch comprises a stylus and a disc comprising 48 concentric grooves. The stylus is controlled to be positioned above the groove to which it is assigned, corresponding to the time to be spoken. The disc is raised to come into contact with the stylus and begins to rotate so that the stylus reads, on the groove, the time to be spoken. The produced sound is thus very short, a few seconds, just enough time for the time to be spoken.

SUMMARY OF THE INVENTION

According to a first aspect, the present invention aims to overcome these disadvantages by proposing a wristwatch which enables the generation of a combination of more than two different sounds, enabling for example music or words to be reproduced within the limited space of a wristwatch.

According to another aspect, the present invention aims to propose a timepiece enabling the generation of sounds during a period of time sufficiently long to allow a piece of music or a speech, for example, to be reproduced.

A further object of the present invention is to enable amplification of the generated sounds to a significant degree, and preferably with limitation, or even cancellation, of unwanted noise.

To this end, the present invention relates to a mechanical wristwatch comprising a case having a middle, a back cover and a bezel, a device for generating vibrations which are intended to produce at least one sound, which is accommodated in said case, a device for amplifying the vibrations generated by said vibration-generating device, and a membrane arranged to receive said amplified vibrations and generate a sound from the inside to the outside of the wristwatch.

According to the invention, the vibration-generating device is a rotary element having a reading surface comprising at least one groove arranged to generate sounds, and the wristwatch comprises means for rotationally driving the rotary element, means for actuating said means for rotationally driving the rotary element, as well as a transmission member arranged to cooperate with the rotary element to transmit, to the amplification device, the vibrations generated by the groove during rotational movement of said rotary element.

In an advantageous manner, the rotary element can comprise at least one groove with several helical coils extending over its reading surface, the transmission member and the rotary element being arranged to be mobile with respect to each other such that the transmission member moves over at least part of the reading surface of the rotary element when said rotary element performs more than one rotation.

Preferably, the vibration-amplifying device comprises a lever pivotally mounted on a support in a pivoting plane non-parallel to the plane of the membrane and arranged to be actuated by the transmission member to transmit the vibrations to said lever, said lever likewise being integral with the membrane at least translationally in the pivoting plane such that a movement of the transmission member, generated by the vibration of said transmission member linked to the rotary element and transmitted to the membrane by way of the lever actuated by said transmission member, generates an amplified movement of said membrane, so as to generate an amplified sound.

Therefore, the present invention enables the generation of combinations of several sounds, and preferably for a time period of several seconds, preferably at least about ten seconds, and more preferably at least one or even several minutes, enabling a piece of music or words to be reproduced, the sound being amplified to a significant degree with limitation, or even cancellation, of unwanted noise, within the restricted space of a mechanical wristwatch.

According to another aspect, the present invention relates to a mechanical timepiece comprising a case having a middle, a back cover and a bezel, a device for generating vibrations which are intended to produce at least one sound, which is accommodated in said case, a device for amplifying the vibrations generated by said vibration-generating device, and a membrane arranged to receive said amplified vibrations and generate a sound from the inside to the outside of the timepiece.

According to this other aspect of the invention, the vibration-generating device is a rotary element comprising a reading surface comprising at least one groove with several helical coils arranged to generate sounds and extending over said reading surface, and the timepiece comprises means for rotationally driving the rotary element, means for actuating said means for rotationally driving the rotary element, as well as a transmission member arranged to cooperate with the rotary element to transmit, to the amplification device, the vibrations generated by the grooves during rotation of said rotary element, the transmission member and the rotary element being arranged to be mobile with respect to each other such that the transmission member moves over at least part of the reading surface of the rotary element when said rotary element performs more than one rotation.

Advantageously, the vibration-amplifying device comprises means for translationally guiding in a plane in parallel with the plane of the membrane, said means arranged to enable said amplification device to follow the movement of the transmission member over at least part of the reading surface of the rotary element when said rotary element performs more than one rotation, and means for mobile coupling between the amplification device and the membrane, said means arranged to enable said amplification device to remain linked with said membrane as it is moving.

Therefore, the present invention enables the generation of combinations of several sounds for a time period of several seconds to several minutes, enabling a piece of music or words to be reproduced using a mechanical timepiece, the sound being amplified to a significant degree with limitation, or even cancellation, of unwanted noise linked to the vibrations of the parts of the movement.

BRIEF DESCRIPTION OF THE DRAWINGS

Other features and advantages of the present invention will become apparent from reading the following detailed description of various embodiments of the invention provided by way of non-limiting example, the description being made with reference to the appended drawings in which:

FIG. **1** is a sectional view of the case of a wristwatch or timepiece according to the invention;

FIG. **2** is a top view of the dial side of a wristwatch according to the invention;

FIG. **3** is a top view of the vibration-amplifying device used in the present invention;

FIG. **4** is a sectional view of the vibration-amplifying device and the membrane;

FIG. **5** is a view of a needle used in the present invention;

FIG. **6** is a sectional view of the membrane and the compression chamber;

FIG. **7** is a view of the means for adjusting the volume of the sound generated by the membrane;

FIGS. **8** and **9** are partial, sectional views respectively showing the volume-adjusting means in the closed position and the open position; and

FIG. **10** is a perspective view of the timepiece according to the invention showing the closure of the openings of the compression chamber outside the sound-generating time.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

According to a first aspect, the present invention relates to a mechanical wristwatch intended to be worn by a user around his wrist. With reference to FIGS. **1**, **2** and **10**, said wristwatch **1** comprises a case **2** having a middle **3**, a back cover **4** closed by a glass **7**, a bezel **5** closed by a glass **6**, a device for generating vibrations which are intended to produce at least one sound mechanically, which is accommodated in said case **2**, a device **16** for amplifying the vibrations generated by said vibration-generating device, and a membrane **18** arranged to receive said amplified vibrations mechanically and generate a sound from the inside to the outside of the wristwatch **1**. The bezel and the middle may be two different parts, or form a single monobloc piece. Similarly, the back cover and the middle may be two different parts, or form a single monobloc piece. The case **2** can also enclose a mechanical timepiece movement allowing the time to be displayed using hands **11** placed between the membrane **18** and the glass **6**. The vibration-generating device and the vibration-amplifying device may be independent of the movement, in the form of a module, unless a link to an alarm is required.

According to the invention, the vibration-generating device is a rotary element **13** having a reading surface comprising at least one groove **15** arranged to generate sounds, and the wristwatch comprises means for rotationally driving the rotary element **13**, means for actuating said means for rotationally driving the rotary element **13**, as well as a transmission member **28** arranged to cooperate with the rotary element **13** to transmit, to the amplification device **16**, the vibrations generated by the groove **15** during rotational movement of said rotary element **13**.

In the following description, the vibration-generating device, the vibration-amplifying device and the membrane are arranged to generate sounds through the dial side of the wristwatch, and more particularly through the glass **6**. Quite obviously, the vibration-generating device, the vibration-amplifying device and the membrane may be arranged similarly but in the reverse manner to generate sounds from the back cover side of the wristwatch, and more particularly through the glass **7**.

The rotary element **13** contains acoustic information to be produced in the form of grooves **15**. These grooves **15** are etched on the rotary element **13** in a suitable manner to generate any type of sound, for example sounds in the form of words (a speech for example) or music (classical music, electronic music, etc.). The rotation of the rotary element **13** enables the generation of vibrations on the membrane **18** via the transmission member **28** and the vibration-amplifying device **16**. According to the first aspect of the invention, the rotary element **13** can comprise one or more grooves each formed of a coil, said coils being concentric in the case of a plurality of grooves, or the rotary element **13** can comprise one or more grooves each formed of several helical coils, the grooves being discontinuous if a plurality of tracks are intended to be formed.

Advantageously, the rotary element **13** is a flat disc mounted in the bezel **5** in parallel with the dial (or with the back cover) of the watch, so as to be rotatable e.g. about the spindle of the hands **11**. This flat disc can even be arranged to form a dial for said wristwatch. The disc can be produced from metal (steel, CuBe, etc.). The grooves **15** are etched mechanically (milling cutter, etching needle, etc.) or by laser. The disc can also be produced by galvanic growth (LIGA [lithography, electroplating and moulding]). It can also be made from silicon, produced by DRIE. Ceramic, a composite or a polymer can also be used.

The rotary element **13** can also be formed by a cylinder, placed around the movement, the grooves being arranged on the inner or outer periphery thereof.

The rotary element **13** must have the lowest possible mass and inertia and be balanced in order to obtain rapid acceleration and must not be influenced by accelerations (gravity, impacts, etc.).

Preferably, the transmission member **28** is a needle arranged to follow at least one groove **15** of the rotary element **13**, said needle being integral with the vibration-amplifying device **16**. Such a needle is shown in FIG. **5**. It must be thin and hard so as not to wear too quickly. It can be produced by steel, CuBe, ceramic, hard metal, ruby, etc. Its end **28***a* for reading the groove can be spherical as shown or conical.

In order to ensure that the sound can be heard many times, the tribologic torque between the rotary element **13** and the transmission member **28** is high because it is the friction of the transmission member **28** in the grooves **15** which generates the vibration. In the event of wear, the acoustic quality would be directly impacted.

Advantageously, the means for rotationally driving the rotary element **13** comprise a spring, of the barrel spring type. The spring is configured to provide the energy necessary for operation of the rotary element **13** to ensure reading one time or multiple times.

The wristwatch also comprises a winding mechanism for the spring. In the case of one-time reading, the spring can be wound by a slide, of the minute repeater type. For a mechanism arranged for multiple-time reading, the spring can be wound by a winding stem, similarly to the barrel of the movement.

Moreover, the wristwatch comprises a regulator arranged to control the rotational speed of the rotary element **13**. This regulator may be of the flywheel type (similar to those used in a minute repeater), of the escapement type with a balance or flexible guiding, of the vane type, or regulated by its own inertia.

The means for actuating the means for rotationally driving the rotary element may be of different types based on the driving means, the reading on one or more rotations, and the function of the generated sounds. Therefore, the means for actuating the means for rotationally driving the rotary element can comprise a slide, of the minute repeater type, enabling winding of the spring as described above as well as the engagement of the rotation of the rotary element **13**. If need be, the slide can also be arranged to bring the transmission member to an initial reading position at the beginning of the reading surface, as described in detail hereinafter.

The means for actuating the means for rotationally driving the rotary element can also comprise a push button arranged to bring the transmission member to an initial reading position at the beginning of the reading surface, as described in detail hereinafter, upon actuation of the push button, then to engage the rotation of the rotary element **13** and enable reading, upon release of said push button.

When the generated sound is used as an alarm, the means for actuating the means for rotationally driving the rotary element can comprise an engagement mechanism linked to a predetermined time and cooperating with the watch movement.

With reference more particularly to FIGS. **1**, **3** and **4**, in order to amplify the sounds produced by the rotary element **13**, the device **16** for amplifying the vibrations generated by said rotary element **13** comprises a lever **20** linked to a support **22** by an articulation along a single axis of rotation, such that said lever **20** is tiltably or pivotally mounted on said support **22** about said axis of rotation, in a pivoting plane non-parallel to the plane of the membrane **18**.

Particularly preferably, the lever **20** is pivotally mounted on its support **22** such that it tilts in a pivoting plane which is perpendicular to the plane defined by the membrane **18**.

Particularly preferably, the lever **20** is pivotally mounted on the support **22** by way of a flexible ball joint **24**. The flexible ball joint makes it possible to change the direction of the vibrations if necessary. A pin which is mounted transversely on the support **22** and constitutes the pivot axis of the lever **20** could also be used.

The lever **20** is arranged to be actuated by the transmission member **28** cooperating with the rotary element **13** to transmit, to the lever **20**, the vibrations generated by the grooves **15**. The transmission member **28**, in this case the needle, is preferably integral with the lever **20**. It is mounted for example by embedding or brazing. To this end, the lever **20** comprises an orifice **29** into which the needle is inserted.

The lever **20** is also integral with the membrane **18** at least translationally in the pivoting plane of the lever **20** in order to be capable of moving said membrane **18** in the pivoting plane of the lever **20** when it tilts. Thus, a movement of the transmission member **28**, generated by the vibration of said transmission member **28** linked to the rotary element and transmitted to the membrane **18** by way of the lever **20** actuated by said transmission member **28**, generates an amplified movement of said membrane **18**, so as to generate an amplified sound. It is the linear movement of the membrane **18** moved by the lever **20** which will generate an amplified sound.

More specifically, the lever **20** comprises a first end **20***a* pivotally mounted on the support **22** in a pivoting plane non-parallel to the plane of the membrane **18** and integral with the transmission member **28** and a second end **20***b* integral with the membrane **18** at least translationally in the pivoting plane of said lever **20**, as described hereinafter.

In the case of a rotary element comprising a single groove formed of a single coil, or a set of grooves with concentric coils (in this case, the wristwatch comprises a mechanism for positioning the transmission member, said mechanism being arranged to position the transition member facing the groove to which it is assigned), the transmission member **28** is fixed relative to the rotary element **13** at the time of reading. The lever **20** is thus translationally fixed and is fixedly connected to the membrane **18**, i.e. it is integral with the membrane **18** rotationally and translationally. The lever **20** can be fixed to the membrane **18** for example by way of a screw screwed into a hole provided on the membrane **18** and into a hole provided on the lever **20**, or by adhesive bonding, welding, etc.

Particularly preferably, the rotary element **13** comprises at least one groove formed of several helical coils extending over its reading surface, enabling reading of the rotary element **13** upon several rotations, which enables the reading time to be increased. In this case, the transmission member **28** and the rotary element **13** are arranged to be mobile with respect to each other such that the transmission member **28** moves over at least part of the reading surface of the rotary element **13** when said rotary element performs more than one rotation. This means that either the transmission member **28** is moved, the rotary element **13** remaining fixed in position, or conversely the transmission member **28** remains fixed in position, the rotary element **13** being moved.

Preferably, in the variant shown, the transmission member **28** is moved, the rotary element **13** remaining fixed in position.

The vibration-amplifying device **16** integral with the transmission member **28** must thus be arranged to be able to follow the movements of said transmission member **28**. A mobile connection between the lever **20** and the membrane **18** must thus be provided.

To this end, the vibration-amplifying device **16** comprises means for translationally guiding in a plane in parallel with the plane of the membrane **18**, said means being arranged to enable said amplification device **16** to follow the movement of the transmission member **28** over at least part of the reading surface of the rotary element when said rotary element **13** performs more than one rotation, and means for mobile coupling between the amplification device **16** and the membrane **18**, said means being arranged to enable said amplification device **16** to remain linked with said membrane **18** as it is moving.

Advantageously, said means for translationally guiding the amplification device comprise a bearing element **50** integral with the case **2** and along which the support **22** of the lever **20** can bear and slide in a plane in parallel with the plane of the membrane **18**. A spring **51** can guarantee good bearing of the support **22** on the bearing element **50**. The bearing element **50** can be mounted on the bezel **5** if the sound is generated on the dial side or on the back cover **4** if the sound is generated on the back cover side, or even on the middle. It is made of a rigid material. It has a length sufficient to enable guiding of the support **22** throughout its movement.

The mobile coupling means comprise a fork **52** integral with the second end **20***b* of the lever and a stud **54** integral with the membrane **18** and having a groove **56** accommodating the two prongs **52***a*, **52***b* of the fork. Therefore, since said fork **52** is engaged with the stud **54**, more specifically in contact with the lateral faces of the stud **54** in its groove **56**, it can slide on either side of said stud **54** in a plane in parallel with the plane of the membrane **18**, whilst being integral with the stud **54** translationally in the pivoting plane, bearing on the stud **54**. Thus, a movement of the needle, generated by the vibration of the needle linked to the rotary element **13** and transmitted to the membrane **18** by way of the lever **20** actuated by said needle, generates an amplified movement of said membrane **18**, so as to generate an amplified sound.

Advantageously, the vibration-amplifying device **16** comprises articulated stabilising arms **58** which are integral with the lever **20**. These arms **58** can be in the form of a pantograph, a tip **58***a* of the pantograph being integral with the support **22** of the lever **20** and the opposite tip **58***b* being integral with the case **2** (bezel, back cover or middle) or with the movement of the wristwatch. The articulations can advantageously be in the form of flexible ball joints **58***c*. Stops **58***d* can be provided to limit the movement of the pantograph.

When reading of the rotary element **13** is effected over several rotations, the wristwatch comprises a mechanism for bringing the transmission member **28** to an initial position corresponding to the beginning of the reading surface of the rotary element **13**, prior to or simultaneously with the actuation of the means for actuating the means for rotationally driving said rotary element **13**. Said mechanism cooperates for example with the slide or push button described above.

The point of actuation of the lever **20** by the transmission member **28**, the pivoting point of the lever **20** and the point where the lever **20** is linked to the membrane **18** are positioned such that a movement of the transmission member **28**, generated by the vibration of said transmission member **28** linked to the rotary element **13** and transmitted to the membrane **18** by way of the lever **20** actuated by said transmission member **28**, generates an amplified movement of said membrane **18** by a mechanical lever arm effect, thereby generating an amplified sound.

Preferably, the flexible ball joint **24** is positioned on the centre of mass of the transmission member/lever/membrane assembly.

Preferably, the point where the lever **20** is linked to the membrane **18** is located at a distance from the centre of said membrane **18** which is less than 40%, preferably less than 20%, of the width of said membrane **18**. Particularly preferably, the point where the lever **20** is linked to the membrane **18** is located at the centre of said membrane **18**, in order to maximise the lever arm effect.

In another variant, the link point can be positioned at the periphery of the membrane, playing on the elasticity of the fixing point and torsion of the membrane.

In the case where the transmission member **28** and the lever **20** are moved, the rotary element **13** remaining fixed in position, the reduction in the amplification of the sound owing to the reduction in the lever arm effect is compensated for by having increased the amplitude of the vibration generated by the groove read by the transmission member **28**.

So as not to damage the rotary element **13** and/or the transmission member **28**, a system of protection in case of impacts is provided.

Advantageously, the vibration-amplifying device, and more particularly the support **22** of the lever **20**, is isolated from the movement and from the frame. To this end, the support **22** of the lever **20** may be mounted on an external part and more particularly on the bezel **5** if the sound is transmitted from the dial side, or on the back cover **4**, if the sound is transmitted through the back cover.

The support **22** of the lever **20** may also be mounted on the movement, or even on the middle **3**, but by way of an isolating element, such as a polymer, in order to ensure the isolation of the support **22** from the movement, the middle and the frame.

When the support **22** of the lever **20** is mounted on one of the elements of the case (bezel **5**, middle **3**, back cover **4**), it is likewise possible to isolate the movement from the remainder of the watch by disposing an isolating device on the movement, between the movement and said element of the case which bears the support **22** of the lever **20**. Such an isolating device may be for example a brass ring associated with a flat gasket or an isolating ring of carbon-filled polyether ether ketone (PEEK). For example, in order to isolate the movement, the latter may be flanged onto a PEEK isolating ring which is itself screwed into the case middle.

When the support **22** of the lever **20** is mounted on the bezel **5**, or on the back cover **4**, said bezel **5**, or said back cover **4**, may be conventionally mounted directly on the middle **3**. However, in order to ensure that little or no unwanted vibration arrives at the middle **3** and therefore at the movement, said bezel **5**, or said back cover **4**, is mounted on the middle **3** by way of an isolating element, such as a polymer.

The membrane **18** is mounted on the back cover **4**, if the sound is transmitted through the back cover, or the bezel **5** if the sound is transmitted from the dial side. Thus, the membrane **18** is also isolated from the movement.

Consequently, the vibration-amplifying device and the membrane are isolated from the movement and the frame such that all the vibrations of the rotary element **13** recovered by the lever **20** are directed and transmitted to the membrane **18**, without transmission to the movement, the frame or the middle.

The lever **20** is preferably made from a very light and very rigid material. For example, the lever **20** may be made of steel, titanium, aluminium, magnesium, composite material, carbon, glass, sapphire and ceramic.

The membrane **18** must be able to vibrate freely. It is preferably made from a very light and rigid material in order have a large dynamic response and not to deform when it moves. The membrane **18** is preferably made of a transparent material. For example, the membrane **18** may be made in the form of a plate of sapphire, mineral glass, Plexiglas, etc. The membrane **18** may thus be constituted by an internal glass disposed under the glass **6** (or the glass **7**, as applicable). The membrane **18** may also be made of composite materials, cellulose, or composite glass of the safety glass type. Quite obviously, the membrane **18** may also be made of a material which is not totally transparent, such as translucent or opaque materials, such as metallic glass, titanium, silicon, ceramic, carbon fibre composite, cellulose or Kevlar.

The membrane **18** must be as rigid and as light as possible. To this end, its thickness is less than 1 mm, preferably less than 0.5 mm and more preferably less than 0.3 mm.

Referring more specifically to FIG. **6**, the membrane **18** is preferably mounted on a suspension to allow it to vibrate freely while ensuring sealing tightness. The suspension may be provided by way of a moulded gasket disposed around the membrane **18** or an O-ring gasket disposed on the periphery on either side of the membrane **18**, between said membrane **18** and the bezel **5** (or the back cover **4**, as applicable). It is likewise possible to texture the membrane **18** to improve its flexibility, for example by way of grooves **38** provided on the periphery of the membrane **18**. It is also possible to play on the thickness of the membrane **18**.

Advantageously, the wristwatch further comprises a compression chamber **40** which is arranged to amplify the sound which has been generated by the movement of the membrane, which has already been amplified a first time, before it emerges to the outside.

The compression chamber **40** is closed by a wall disposed downstream of the membrane **18** and having at least one opening **42** directed towards the outside of the timepiece. The dimensions of the opening **42** are selected in relation to the surface area of the membrane **18** as a function of the desired sound amplification.

Preferably, when the membrane **18** is disposed on the bezel (or back cover **4**), said wall is constituted by the glass **6** (or glass **7**). Said glass **6** or **7** must be strong and scratchproof. It is preferably made of sapphire.

The wall disposed downstream of the membrane **18** makes it possible to generate a certain volume of air in the compression chamber **40**. Movement of the membrane **18** compresses this air. The ratio between the surface area of the moving membrane **18** which defines the volume of air and the size of the openings will make it possible to achieve a second amplification of the sound.

The opening **42** may take various forms: a hole in the centre of the wall, in this case the glass **6**, a plurality of lateral holes, a multitude of micro-holes, etc. The shape of the outlet opening may be conical to amplify the sound further. Advantageously, the opening **42** may take the form of one or more longitudinal grooves extending around the middle, parallel to the plane of the movement, and obtained by leaving a space for example between the back cover **4** and the middle **3** if the membrane **18** is disposed on the back cover **4**, as shown in FIG. **10**. Such lateral openings **42** advantageously make it possible to redirect the sound generated by the membrane **18** onto the wristwatch wearer's arm, giving said wearer the sensation of even further amplified sound.

With reference to FIGS. **7** to **9**, the wristwatch advantageously comprises adjustable closure means **44** of the opening **42** of the compression chamber **40** which are arranged to make it possible to adjust the volume of sound generated by the membrane **18**. These closure means **44** are arranged to be movable between a closed position in which the opening **42** is entirely closed and an open position in which the opening **42** is at its largest, the intermediate positions allowing the opening **42** to be blocked to a greater or lesser extent in order to adjust the sound volume. For example, when the opening **42** is a lateral opening, the closure means **44** may comprise a rotary element such as a ring, mounted rotatably on the bezel **5** (or back cover **4** if applicable), and arranged to rotate between a closed position in which the opening **42** is entirely blocked, as shown in FIG. **8**, and an open position in which the opening **42** is at its largest, as shown in FIG. **9**.

Advantageously, the opening **42** may be protected by an air-permeable and water-proof filter, in order to prevent impurities from entering the compression chamber **40**. Such filters are known to a person skilled in the art.

Advantageously, and with reference to FIG. **10**, in which the devices and the membrane are arranged on the back cover side, the sound being generated on the back cover side, the wristwatch may comprise a mechanism for closing the opening **42** of the compression chamber **40** arranged to close said opening **42** once the sound has been generated, for example outside the striking time. Such a mechanism makes it possible to limit the ingress of impurities when the vibration-generating device is not in use. In particular, when the opening **42** is a lateral opening, the closing mechanism may comprise a closing member **46**, such as a ring, controlled by the striking-mechanism so as to close the opening **42** automatically once striking is complete. Such a mechanism is described, for example, in patent CH 704 940, which is incorporated by reference.

When the membrane **18** is disposed on the back cover **4** and sound is generated from the back cover side, it is also possible to provide a reflector, arranged to allow the generated sound to rise up the side of the watch. Such a reflector is for example positioned on the outside of the back cover, and has conical outer walls surrounding the back cover and directed towards the watch.

The wristwatch according to the invention enables the generation, using the rotary element comprising etched grooves, of combinations of several sounds, and preferably for a time period of several seconds to several minutes, enabling a piece of music or words to be reproduced, within the restricted space of a mechanical wristwatch.

Furthermore, the wristwatch according to the invention enables significant sound amplification: the ratio of the first amplification performed by way of the lever **20** is of the order of 10 to 16. Furthermore, due to the ratio between the moving surface area of the membrane **18** and the opening **42** (surface area ratio of 20 to 30), the second amplification ratio is of the order of 1.5. The total amplification ratio achieved by the watch according to the invention is such that a sound greater than 80 dB, or even 90 dB, may be obtained, as compared to a standard minute-repeater sound of 60 dB.

Furthermore, the construction and mounting of the amplification device make it possible to isolate the rotary element and the membrane from the movement, the frame and the middle, such that all the vibrations are directed onto the membrane by way of the lever without passing via the movement. Little, or even no, unwanted sound generally associated with the vibrations of movement components is thus produced.

Moreover, since the amplification device with the membrane is integral with the back cover or the bezel, the movement can be easily accessed simply once the back cover or the bezel is just removed.

Finally, the construction of the amplification device with the membrane is such that the sealing tightness of the case has no impact on sound volume, unlike all other minute-repeater principles, for example. This is because the membrane with its suspension makes the system leaktight with the exception of the compression chamber **40**. The latter may be made leaktight by a filter system.

According to another aspect, the present invention relates more generally to a mechanical timepiece **1** comprising a case **2** having a middle **3**, a back cover **4** and a bezel **5**, a device for generating vibrations which are intended to produce at least one sound mechanically, which is accommodated in said case **2**, a device **16** for mechanically amplifying the vibrations generated by said vibration-generating device, and a membrane **18** arranged to receive said mechanically amplified vibrations and generate a sound from the inside to the outside of the timepiece, the vibration-generating device being a rotary element **13** having a reading surface comprising at least one groove **15** formed of several helical coils, said groove being arranged to generate sounds and extending over said reading surface. Said timepiece comprises means for rotationally driving the rotary element **13**, means for actuating said means for rotationally driving the rotary element **13**, and a transmission member **28** arranged to cooperate with the rotary element **13** to transmit, to the amplification device **16**, the vibrations generated by the grooves **15** during rotation of said rotary element **13**, the transmission member **28** and the rotary element **13** being arranged to be mobile with respect to each other such that the transmission member **28** moves over at least part of the reading surface of the rotary element **13** when said rotary element **13** performs more than one rotation.

Advantageously, the vibration-amplifying device **16** comprises means for translationally guiding in a plane in parallel with the plane of the membrane **18**, said means being arranged to enable said amplification device **16** to follow the movement of the transmission member **28** over at least part of the reading surface of the rotary element **13** when said rotary element **13** performs more than one rotation, and means for mobile coupling between the amplification device **16** and the membrane **18**, said means being arranged to enable said amplification device **16** to remain linked with said membrane **18** as it is moving.

Advantageously, the vibration-amplifying device **16** comprises a lever **20** having a first end **20***a* pivotally mounted on a support **22** in a pivoting plane non-parallel to the plane of the membrane **18** and integral with the transmission member **28** so as to be able to be actuated by said transmission member **28** for transmitting, to the lever **20**, the vibrations generated by the rotary element **13**, said means for translationally guiding the amplification device comprise a bearing element **50** integral with the case **2** and along which the support **22** of the lever **20** can bear and slide in a plane in parallel with the plane of the membrane **18**. The mobile coupling means comprise a fork **52** integral with the second end **20***b* of the lever **20** and a stud **54** integral with the membrane **18**, said fork **52** being engaged with the stud **54** in order to slide on either side of said stud **54** in a plane in parallel with the plane of the membrane **18**, whilst being integral with the stud **54** translationally in the pivoting plane. Thus, a movement of the needle, generated by the vibration of the needle linked to the rotary element **13** and transmitted to the membrane **18** by way of the lever **20** actuated by said needle, generates an amplified movement of said membrane **18**, so as to generate an amplified sound.

Advantageously, the vibration-amplifying device **16** comprises articulated stabilising arms **58** which are integral with the lever **20**.

The timepiece further comprises a mechanism for bringing the transmission member **28** to an initial position corresponding to the beginning of the reading surface of the rotary element **13**, prior to or simultaneously with the actuation of the means for actuating the means for rotationally driving the rotary element **13**.

The transmission member is advantageously a needle arranged to follow the grooves **15** of the rotary element **13**, said needle being integral with the vibration-amplifying device **16**.

Preferably, the rotary element **13** is a flat disc or a cylinder. It can be a flat disc arranged to form a dial for said timepiece.

Advantageously, the grooves **15** provided on the rotary element are arranged to generate sounds in the form of words or music.

Advantageously, the timepiece comprises a compression chamber **40** arranged to amplify the sound which has been generated by the movement of the membrane **18** before it emerges to the outside. Said compression chamber **40** can be closed by a wall disposed downstream of the membrane **18** and having at least one opening **42**, the dimensions of which are selected in relation to the surface area of the membrane **18** as a function of the desired sound amplification.

The construction details and the advantages of the wristwatch according to the first aspect of the invention, in particular the details relating to the vibration-amplifying device, the means for translationally guiding and the mobile coupling means, similarly apply to the timepiece according to the second aspect of the invention.

The invention claimed is:

1. A mechanical wristwatch comprising a case having a middle, a back cover and a bezel, a vibration-generating device for generating vibrations which are intended to produce at least one sound, which is accommodated in said case, a vibration-amplifying device for amplifying the vibrations generated by said vibration-generating device, and a membrane arranged to receive said amplified vibrations and generate a sound from the inside to the outside of the wristwatch, wherein the vibration-generating device is a rotary element having a reading surface comprising at least one groove extending over the reading surface of the rotary element, said at least one groove being arranged to generate sounds, and wherein the wristwatch comprises means for rotationally driving the rotary element, means for actuating said means for rotationally driving the rotary element, and a transmission member arranged to cooperate with the rotary element to transmit, to the vibration-amplifying device, the vibrations generated by the groove during rotational movement of said rotary element,

wherein the transmission member and the rotary element are arranged to be mobile with respect to each other such that the transmission member moves over at least part of the reading surface of the rotary element when said rotary element performs more than one rotation, and wherein the vibration-amplifying device comprises a lever pivotally mounted on a support in a pivoting plane non-parallel to the plane of the membrane and arranged to be actuated by the transmission member to transmit the vibrations to said lever, said lever likewise being integral with the membrane at least translationally in the pivoting plane such that a movement of the transmission member, generated by the vibration of said transmission member linked to the rotary element and transmitted to the membrane by way of the lever actuated by said transmission member, generates an amplified movement of said membrane by a mechanical lever arm effect, so as to generate an amplified sound.

2. The wristwatch according to claim 1, wherein the means for rotationally driving the rotary element comprise a spring and a winding mechanism for said spring.

3. The wristwatch according to claim 1, further comprising a regulator arranged to control the rotational speed of the rotary element.

4. The wristwatch according to claim 1, wherein the means for actuating the means for rotationally driving the rotary element comprise a slide, a push button or an engagement mechanism linked to a predetermined time and cooperating with a watch movement.

5. The wristwatch according to claim 1, further comprising a compression chamber which is arranged to amplify the sound which has been generated by movement of the membrane before the sound emerges to the outside.

6. The wristwatch according to claim 5, wherein the compression chamber is closed by a wall disposed downstream of the membrane and having at least one opening, the dimensions of which are selected in relation to the surface area of the membrane as a function of the desired sound amplification.

7. The wristwatch according to claim 6, wherein the lever has a first end pivotally mounted on the support in a pivoting plane non-parallel to the plane of the membrane and integral with the transmission member, wherein said means for translationally guiding the vibration amplifying device comprise a bearing element integral with the case and along which the support of the lever can bear and slide in a plane in parallel with the plane of the membrane, and wherein the mobile coupling means comprise a fork integral with a second end of the lever and a stud integral with the membrane, said fork being engaged with the stud in order to slide on either side of said stud in a plane in parallel with the plane of the membrane, whilst being integral with the stud translationally in the pivoting plane.

8. The wristwatch according to claim 7, wherein the vibration-amplifying device comprises articulated stabilising arms which are integral with the lever.

9. The wristwatch according to claim 1, wherein the vibration-amplifying device comprises means for translationally guiding in a plane in parallel with the plane of the membrane, said means being arranged to enable said vibration-amplifying device to follow the movement of the transmission member over at least part of the reading surface of the rotary element when said rotary element performs more than one rotation, and means for mobile coupling between the vibration-amplifying device and the membrane, said means being arranged to enable said vibration-amplifying device to remain linked with said membrane as the membrane is moving.

10. The wristwatch according to claim 1, further comprising a mechanism for bringing the transmission member to an initial position corresponding to a beginning of the reading surface of the rotary element, prior to or simultaneously with an actuation of the means for actuating the means for rotationally driving the rotary element.

11. The wristwatch according to claim 1, wherein the transmission member is a needle arranged to follow the at least one groove of the rotary element, said needle being integral with the vibration-amplifying device.

12. The wristwatch according to claim 1, wherein the rotary element is a flat disc or a cylinder.

13. The wristwatch according to claim 12, wherein the rotary element is a flat disc.

14. The wristwatch according to claim 1, wherein the groove provided on the rotary element is arranged to generate sounds in the form of words or music.

15. A mechanical timepiece comprising a case having a middle, a back cover and a bezel, a vibration-generating device for generating vibrations which are intended to produce at least one sound, which is accommodated in said case, a vibration-amplifying device for amplifying the vibrations generated by said vibration-generating device, and a membrane arranged to receive said amplified vibrations and generate a sound from the inside to the outside of the timepiece, wherein the vibration-generating device is a rotary element having a reading surface comprising at least one groove arranged to generate sounds and extending over said reading surface, and wherein the timepiece comprises means for rotationally driving the rotary element, means for actuating said means for rotationally driving the rotary element, and a transmission member arranged to cooperate with the rotary element to transmit, to the vibration-amplifying device, the vibrations generated by the grooves during rotation of said rotary element, the transmission member and the rotary element being arranged to be mobile with respect to each other such that the transmission member moves over at least part of the reading surface of the rotary element when said rotary element performs more than one rotation, and

wherein the vibration-amplifying device comprises a lever pivotally mounted on a support in a pivoting plane non-parallel to the plane of the membrane and arranged to be actuated by the transmission member to transmit the vibrations to said lever, said lever likewise being integral with the membrane at least translationally in the pivoting plane such that a movement of the transmission member, generated by the vibration of said transmission member linked to the rotary element and transmitted to the membrane by way of the lever actuated by said transmission member, generates an amplified movement of said membrane by a mechanical lever arm effect, so as to generate an amplified sound.

16. The timepiece according to claim 15, wherein the vibration-amplifying device comprises means for translationally guiding in a plane in parallel with the plane of the membrane, said means being arranged to enable said support of the lever to follow the movement of the transmission member over at least part of the reading surface of the rotary element when said rotary element performs more than one rotation, and means for mobile coupling between the lever and the membrane, said means being arranged to enable said lever to remain linked with said membrane as the membrane is moving.

17. The timepiece according to claim 16, wherein the vibration-amplifying device comprises a lever having a first end pivotally mounted on a support in a pivoting plane non-parallel to the plane of the membrane and integral with the transmission member, wherein said means for translationally guiding the vibration amplifying device comprise a bearing element integral with the case and along which the support of the lever can bear and slide in a plane in parallel with the plane of the membrane, and wherein the mobile coupling means comprise a fork integral with the second end of the lever and a stud integral with the membrane, said fork being engaged with the stud in order to slide on either side of said stud in a plane in parallel with the plane of the membrane, whilst being integral with the stud translationally in the pivoting plane.

18. The timepiece according to claim 17, wherein the vibration-amplifying device comprises articulated stabilising arms which are integral with the lever.

19. The timepiece according to claim **15**, further comprising a mechanism for bringing the transmission member to an initial position corresponding to a beginning of the reading surface of the rotary element, prior to or simultaneously with an actuation of the means for actuating the means for rotationally driving the rotary element.

20. The timepiece according to claim **15**, wherein the transmission member is a needle arranged to follow the grooves of the rotary element, said needle being integral with the vibration amplifying device.

21. The timepiece according to claim **15**, wherein the rotary element is a cylinder.

22. The timepiece according to claim **15**, wherein the grooves provided on the rotary element are arranged to generate sounds in the form of words or music.

23. The timepiece according to claim **15**, further comprising a compression chamber which is arranged to amplify the sound which has been generated by the movement of the membrane before the sound emerges to the outside.

24. The timepiece according to claim **23**, wherein the compression chamber is closed by a wall disposed downstream of the membrane and having at least one opening, the dimensions of which are selected in relation to the surface area of the membrane as a function of the desired sound amplification.

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